

MTH1001 Mathematical Structures 2025–26, Term 1

Problem Sheet 5

This sheet is for discussion in the week 12 tutorials (Wednesday 10th December). Be ready to discuss (at least) Questions A4(iii)(iv), A6, A7(ii), B3(iii)–(vi), B4 in the tutorial.

Your answers should use mathematical notation correctly and be written in logically correct sentences, including words such as "let", "if", "then". You should introduce any variables that occur in your argument.

SECTION A

These are fairly routine questions. There will be some questions on the January exam of a similar difficulty to these, and some more challenging questions.

A1. Let G be a group with operation $*$. Show that G has only one identity element (if e_1 and e_2 are identity elements, what is $e_1 * e_2$?) Also show that each element has only one inverse (if b_1 and b_2 are inverses of a , what is $b_1 * a * b_2$?)

A2. Let G be a group with operation $*$. Let $a, b \in G$. Prove that the equation

$$a * x = b$$

has a unique solution for $x \in G$ and that the equation

$$y * a = b$$

has a unique solution for $y \in G$. Must x equal y ?

A3. Let G be a group written in multiplicative notation.

(i) If $a^2 = a$ for some $a \in G$, prove that $a = e$, the identity.

(ii) For $b, c \in G$, prove that the inverse of bc is $c^{-1}b^{-1}$.

A4. For each of the following sets G , determine whether G is a group under the given operation $*$, justifying your answers. If it is a group, find the identity element and the inverse of each $g \in G$. If it is not a group, indicate how at least one of the group axioms fails.

(i) $G = \mathbb{Z}$, $a * b = a - b$;

(ii) $G = \mathbb{R}^\times$, $a * b = \frac{1}{2}ab$;

- (iii) $G = \mathbb{R}$, $a * b = \min(a, b)$, the minimum of a and b ;
 - (iv) $G = \{x \in \mathbb{R} : x \neq -1\}$, $x * y = x + y + xy$.
- A5. Let D_{12} denote the group of symmetries of a regular hexagon. Describe each of its elements, in particular giving the order of each element.
- A6. Write down the conditions for a subset H of a group G to be a subgroup. Use these to show that if H_1 and H_2 are subgroups of G then so is their intersection $H_1 \cap H_2$.
- A7. In each of the following cases, list the elements of the group $\mathbb{Z}_n^\times$, and find the order and the inverse of each element:

- (i) $n = 9$; (ii) $n = 10$; (ii) $n = 12$.

SECTION B

These are more challenging questions. There may be questions on the January exam of similar difficulty to these.

- B1. (i) Let G be a finite group (written multiplicatively) and let $g \in G$ have order n . Show that, for any positive factor d of n , the element g^d has order n/d .
- (ii) Now suppose that g has order n and let $k \in \mathbb{N}$ be arbitrary. Find and prove an expression for the order of g^k in terms of n and k .
- B2. Recall that S_n is the group of all permutations of the set $\{1, 2, \dots, n\}$ and that A_n is the subgroup of S_n consisting of all even permutations in S_n . Let $n \geq 2$ and let

$$B_n = \{\sigma \in S_n : \sigma \text{ is an odd permutation}\}.$$

Show that the function $f : A_n \rightarrow B_n$ given by $f(\sigma) = (12)\sigma$ is a bijection. Hence find the order of the group A_n .

- B3. In each case determine whether or not $\phi : G \rightarrow H$ is a homomorphism. (The operations on G and H are the standard ones in each case.) Justify your answers.
- (i) $G = \mathbb{Z}$, $H = \mathbb{Z}$, $\phi(n) = -5n$;
 - (ii) $G = \mathbb{R}$, $H = \mathbb{R}$, $\phi(x) = x^3$;

- (iii) $G = \mathbb{R}^\times, H = \mathbb{R}, \phi(x) = x^5$;
- (iv) $G = \mathbb{R}^\times, H = \mathbb{R}^\times, \phi(x) = x^7$;
- (v) $G = S_4, H = S_4, \phi(\sigma) = \sigma^{-1}$;
- (vi) $G = \mathbb{R}^\times, H = \mathbb{Q}^\times, \phi(x) = \begin{cases} 1 & \text{if } x > 0; \\ -1 & \text{if } x < 0. \end{cases}$

- B4. Let G be a group (written multiplicatively) and let $h \in G$. Let $\phi : G \rightarrow G$ be the function given by $\phi(g) = hgh^{-1}$ for all $g \in G$. Show that ϕ is an isomorphism of groups. What is its inverse isomorphism?
- B5. We can consider symmetry groups in three dimensions as well as two dimensions.

Let C be a cube, and let G be the set of all rotations taking C to itself. Certainly G contains the identity operation which I'll call I . Also C has various rotational axes of symmetry. It has three axes of symmetry joining the centre of a face to that of the opposite face, and four axes of symmetry joining a vertex to the opposite vertex. How many rotational symmetries are there about these axes, and what are the orders of these rotations?

Are these all of the rotational symmetries, or are there more? If there are more, describe the other rotations. In any case find the order of G .

SECTION C: CHALLENGE PROBLEMS

- C1. Complete the following group multiplication table.

	s	t	u	v	w	x	y	z
s	s							
t		s						
u		v	t		y			
v				t				
w		x			t		u	
x						t		
y		z	w				t	
z								t

Some hints: Identify the identity. Find the order of each element. Identify the powers of each element. Find the inverse of each element.

C2. (i) Let S be a set and let $*$ be a binary operation on S so that the following properties hold:

- $(x * y) * z = x * (y * z)$ for all $x, y, z \in S$;
- there is an element $e \in S$ such that $e * x = x$ for all $x \in S$;
- for each $x \in S$, there is an element $x' \in S$ such that $x' * x = e$.

Prove that S is a group with the operation $*$.

(ii) Let T be a set and let $\diamond$ be a binary operation on T so that the following properties hold:

- $(x \diamond y) \diamond z = x \diamond (y \diamond z)$ for all $x, y, z \in T$;
- there is an element $e \in T$ such that $e \diamond x = x$ for all $x \in T$;
- for each $x \in T$, there is an element $x' \in T$ such that $x \diamond x' = e$.

Does the operation $\diamond$ necessarily make T into a group? Give a proof or a counter-example.